

Cross-dialectal Formulations of the Occurrence Conditions of *ragyō-godanka* in Japanese Dialects, Focusing on Stem-final Vowels, the Stem Mora Count, and Suffixes: A Preliminary Report

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Ragyō-godanka is a phenomenon in various Japanese dialects, in which a vowel-final verb stem behaves morphologically the same as a consonant-final verb stem ending in /r/ (e.g., *miran* /mir-an/ ‘do not see’; Shiba Village dialect, Miyazaki Prefecture). This paper describes the dialectal variation of the verb stem and suffix conditions under which *ragyō-godanka* occurs and then formulates these variations as implicational hierarchies with regard to stem-final vowels, stem mora counts, and suffixes. Finally, it elucidates the motivation for the value ordering of these hierarchies as reflecting the frequencies of word forms.

Keywords: verb morphology, analogical change, verb stem, inflectional suffix, frequency

1 Introduction

This paper presents an intragenetic typological survey of micro-variation of *ragyō-godanka* (henceforth, RG) in Japanese dialects, a phenomenon in which a vowel-final verb stem (or *ichidan/nidan* stem) morphologically behaves the same as a consonant-final verb stem ending in /r/ (or *ragyō-godan* stem) in modern Japanese dialects (i.e., Tohoku, Hokuriku, Tokai, Kinki, Umpaku, Chugoku-Shikoku, and Kyushu dialects).¹

Let us take the RG of the Shiba Village dialect (Miyazaki Prefecture; author data) as an initial approximation. Table 1 is a partial set of the inflectional paradigm of the V-final stem ‘see’. Here, the past form (E) *mita* is formed by adding the past suffix /-ta/ to the vowel-final stem /mi-/. This past form is the same as the past form (G) *huteta* of a V-stem ‘discard’, which alternates its final vowels depending on the suffix it takes (henceforth, *e/u* stem).² On the other hand, for the negative non-past form (C) *miran*, unlike the case of (F) *huten* ‘do not discard’, the r-final stem /mir-/ is followed by a negative non-past suffix /-an/. In other words, the V-stem ‘see’ behaves the same as (G) *e/u*-stem ‘discard’ in (E) *mita* and the same as (A) r-stem ‘shave’ in

¹ From a diachronic perspective, RG is considered as a phenomenon in which a vowel-final stem changes into a r-stem when followed by a specific suffix. In the ‘see’ row of Table 1, this refers to the change of the negative non-past form (D) *min* /mi-n/ to (C) *miran* /mir-an/.

In fact, RG forms of Japanese dialects, as which they are regarded in previous studies, differ in terms of synchronic morphological structure and diachronic change. In short, in addition to the case mentioned above, there are also cases where a vowel-initial suffix behaves morphologically the same as an r-initial suffix (mentioned in §3.3). From a diachronic perspective, this is a phenomenon in which a vowel-initial suffix is changed to a r-initial suffix. In this paper, the two are jointly referred to as RG as in previous studies.

² In this paper, the following class names for vowel-final verb stems are used. For stem-final vowels, the /i/ (so-called *kami-ichidan* ‘upper monograde’) is the *i*-stem, alternating between /i/ ~ /u/ (so-called *kami-nidan* ‘upper bigrade’) is the *i/u*-stem, the /e/ (so-called *shimo-ichidan* ‘lower monograde’) is the *e*-stem, and alternating between /e/ ~ /u/ (so-called *shimo-nidan* ‘lower bigrade’) is the *e/u*-stem. The stem classes are set from synchronic data in each dialect.

(C) *miran*. RG refers to the phenomenon in (C) *miran*, where a vowel-final stem behaves morphologically the same as a consonant-stem ending in /r/.

Table 1: Verb forms of the dialect in Shiba Village (Miyazaki Prefecture; author data)³

		stem	negative non-past		past
Consonant	r	togir- ‘shave’	(A) <i>togiran</i> /togir-an/	(B) <i>togitta</i> /togir-ta/	
Vowel	i	mi(r)- ‘see’	(C) <i>miran</i> /mir-an/	(D) <i>min</i> /mi-n/	(E) <i>mita</i> /mi-ta/
	e/u	hute- ‘discard’	(F) <i>huten</i> /hute-n/		(G) <i>huteta</i> /hute-ta/

This study analyzes data from 60 dialects that exhibit RG. While this is not a comprehensive analysis of all cases of RG in Japanese dialects, it should help to paint a picture of some of the general trends of this phenomenon. It will be shown that, as previous works suggest (e.g., Kobayashi 1996, among others), RG exhibits considerable dialectal variation; however, the observed variation is not random but rather severely constrained, following a certain pattern that has never previously been discussed in the literature. The purpose of the present study is to offer a model with which the pattern is explicitly and exhaustively described and to provide an explanation as to why such a pattern in RG emerges in Japanese dialects. The model takes the form of a set of hierarchies, which is schematically shown in (1) and (2).

- (1) Verb stem hierarchies
 - a. the stem-final vowel: /i/ > /e/
 - b. stem mora count: 1 > 2
- (2) Suffix hierarchies Volitional > Negative non-past > Past
 - a. Eastern-type: Imperative = Volitional
 - b. Western-type: Volitional > Imperative > Negative non-past
 - c. Kyushu-type: Volitional > Negative non-past > Imperative > Past

The hierarchy of the stem-final vowel states that i-final stems (the value ‘/i/’) are more likely to undergo RG than e-final stems (‘/e/’). Hence, it is predicted that if there is a dialect where an e-final stem undergoes RG, then an i-final stem must also undergo RG. The hierarchy of the stem mora count states that monomoraic stems (‘1’) are more likely to undergo RG than bimoraic stems (‘2’). The suffix-type hierarchies state that the likelihood of RG occurrence can be estimated with respect to the inflectional suffixes to which the stem is attached and that there are three different versions of hierarchy depending on the three dialectal types (see §4 for details).

Crucially, the set of hierarchies is not a language-specific and accidental collection of constraints. In fact, it reflects a cross-linguistic mechanism whereby language change is heavily influenced by frequency. As will be discussed in detail in §5, RG can be regarded as a kind of analogical change. In usage-based theory, such analogical change is more likely to occur for

³ In this paper, data are represented phonemically. The underlying representations are indicated by / /, and the surface representations by italics.

low-frequency word forms and less likely to occur for high-frequency forms (Bybee 2015, etc.). According to this viewpoint, a hypothesis arises that the hierarchies (1) and (2), which indicate the likelihood of occurrence of RG, reflect the frequency of verb forms, that is, the values in the hierarchies are ordered from left to right by increasing frequency. For this paper, a corpus survey was conducted, and the results show that this hypothesis is indeed valid.

The structure of the present paper is as follows. In §2, previous studies of RG and its dialectal variation are overviewed, and problems with each are pointed out. In §3, an alternative survey method for overcoming these problems is shown. In §4, formulations are presented as implicational hierarchies based on three aspects: the stem-final vowel, the stem mora count, and the succeeding suffixes. §5 explains the existence of these hierarchies based on frequency.

2 Previous studies of RG and its dialectal variation

Kobayashi (1996) was the first study to deal with the dialectal variation of RG in Japanese dialects. Kobayashi (1996) examined “The Grammar Atlas of Japanese Dialects (GAJ)” (National Institute for Japanese Language and Linguistics [NINJAL] 1991, 1993). The GAJ includes survey items for the verb forms shown in Table 2. The vertical axis indicates the six verb stems (i.e., ‘see,’ ‘get up,’ ‘sleep,’ ‘open,’ ‘come,’ and ‘do’). The horizontal axis indicates the five succeeding suffixes (i.e., negative non-past (NEG), past (PST), volitional (VOL), imperative (IMP), and causative (CAUS) suffixes). ‘ND’ indicates that the combination of the verb stems and succeeding suffixes are not listed in the survey item and are not surveyed. Kobayashi (1996) used the GAJ preliminary investigation to fill in the missing data.

Table 2: Conditions in which RG occurs and GAJ (ND: no data)

verb stem	negative	past	volitional	imperative	causative (non-past)
‘see’	#74 ‘not see’	ND	ND	#86 ‘Look!’	ND
‘get up’	#72 ‘not get up’	ND	#106 ‘will get up’	#85 ‘Get up!’	ND
‘sleep’	#79 ‘not sleep’	ND	#108 ‘will sleep’	ND	ND
‘open’	#77 ‘not open’	ND	#107 ‘will open’	#87 ‘Open!’	#119 ‘make X open’
‘come’	#83 ‘not come’	ND	#110 ‘will come’	#90 ‘Come!’	#120 ‘make X come’
‘do’	#84 ‘not do’	ND	#111 ‘will do’	#91 ‘Do!’	#121 ‘make X do’

Kobayashi (1996) counted the number of RG forms in the GAJ and its preliminary investigation data. In (3), for each verb stem, the number of RG forms for NEG, PST, VOL, and IMP is counted, and the relative frequencies of the forms are shown. In (4), for each suffix, the number of RG forms for the six verb stems is counted, and the relative frequencies of the forms are shown. I have added the number for each word form in square brackets according to Kobayashi’s (1996) survey.

- (3) The number of RG forms for each verb stem
 ‘get up’ $\cong$ ‘see’ $\cong$ ‘sleep’ $>$ ‘open’ $>$ ‘do’ $>$ ‘come’
 [87] [89] [86] [36] [12] [4]
- (4) The number of RG forms for each suffix
 causative $>$ volitional $\cong$ imperative $>$ negative $>$ past
 [256] [121] [119] [101] [10]

A similar claim based on the count data is made for each dialect area: Kinki (Yamaoka 2016), Shikoku (Matsumaru 2001), and Kyushu (Jinnouchi 1989). The number of RG forms shown in (3) and (4) and the studies in the literature making similar claims indicate that verb stems and suffixes are important factors affecting the likelihood of RG occurrence and non-occurrence in Japanese dialects.

In these studies, data from multiple dialects in a certain dialect area or from the whole of Japanese dialects are lumped together. Therefore, it is not clear whether RG occurs or not in certain verb stems and suffixes in each dialect. For example, in Kokonoe Town dialect (Oita Prefecture; Itoi 1964), for the verb stem, RG occurs in /mi-/ ‘see’ and /ne-/ ‘sleep’, but not in /oki-/ ‘get up’ or /ake-/ ‘open’. This difference cannot be captured by (3), in which the number of forms of bimoraic stems such as ‘get up’ is about the same as that of forms of monomoraic stems such as ‘see’ and ‘sleep’. Likewise, the condition of the suffixes in Kokonoe Town dialect, in which RG occurs when the VOL and NEG suffix follows the stem, cannot be captured by (4).

Another problem in previous works dealing with RG (Kobayashi 1996; Yanaike 2006; Hikosaka 2017; Konishi 2017) lies in the fact that they all make use of the GAJ as a sole or main source of the data. The GAJ is not specifically designed for RG and does not present exhaustive data on verb stems and suffixes related to RG (see the ‘ND’ in Table 2). As will be shown in detail in §4, two major factors may affect the likelihood of RG occurrence, namely the verb stems and the type of inflectional suffixes (see (1) and (2) in §1). This fact could never be captured with the GAJ, which does not pay attention to these factors in a systematic way and thus fails to list all combinations required for an exhaustive study of RG.

The lack of systematicity is accompanied by another fatal drawback of the GAJ. It does not provide negative evidence, that is, information regarding whether a given form allows or disallows RG in a given dialect. Since the GAJ plots the data based on the speaker’s response of which form or forms are appropriate or acceptable in a given context, there is no way to know which form(s) the speaker regarded as impossible. The GAJ data in Table 2 sometimes contain “textual notes” providing negative evidence, but these are limited in number and not exhaustive. This scarcity of negative evidence has serious consequences for our efforts to make formulations about RG, as it prohibits the formulation of constraints, that is, the conditions under which RG never occurs (see §3.1 for the survey methodology).

In this paper, contrary to previous studies, the conditions under which RG occurs in each dialect are investigated, and formulations are made based on these conditions in order to overcome the shortcomings mentioned above. By contrasting the Shiba Village dialect (§1),

the Kokonoe Town dialect (Oita Prefecture; Itoi 1964), and the Izumo City dialect (Shimane Prefecture; author data), the dialectal variations are illustrated (Table 3).

Table 3: Comparison of the conditions of RG (+: RG occurs, -: RG doesn't occur)

verb	stem	stem	Shiba (author data)				Kokonoe (Itoi 1964)				Izumo (author data)			
stem	final	morae	VOL	NEG	IMP	PST	VOL	NEG	IMP	PST	VOL	IMP	NEG	PST
mi- 'see'	/i/	1	+	+	+	-	+	+	-	-	+	+	+	-
oki- 'get up'	/i/	2	+	+	+	-	-	-	-	-	+	+	+	-
ne- 'sleep'	/e/	1	+	+	+	-	+	+	-	-	+	+	+	-
ake- 'open'	/e/	2	-	-	-	-	-	-	-	-	+	+	+	-

According to my own and other previous studies, RG is not a lexical phenomenon but is subject to various conditions in each dialect. In each dialect, three aspects condition the occurrence of RG: the stem-final vowel, the stem mora count, and the succeeding suffixes. According to these criteria, the vowel-final stems can be analyzed as a monomoraic i-final for /mi-/ 'see', a bimoraic i-final stem for /oki-/ 'get up', a monomoraic e-final for /ne-/ 'sleep', and a bimoraic e-final for /ake-/ 'open', respectively. In the survey I conducted, several verb stems with the same conditions for the stem-final vowel and the stem mora count were investigated. For example, for a monomoraic i-final stem, /mi-/ 'see', /ki-/ 'wear', /ni-/ 'boil', etc. were investigated. The data show that verb stems with the same conditions exhibit the same morphological behavior. In some data from previous studies such as Itoi (1964), the same behavior is explicitly shown for multiple verb stems with the same conditions. Thus, it is likely that the conditions of the stem-final vowel and the stem mora count rather than lexical conditions are involved in RG.⁴ In the dialects in Table 3, there are also conditions under which RG occurs. In the Shiba Village dialect, RG does not occur in bimoraic stems with the stem-final vowel /e/, regardless of the following suffix. Except for bimoraic e-final stems, RG occurs when the NEG, VOL, and IMP suffixes follow. In the Kokonoe Town dialect (Itoi 1964), RG occurs in monomoraic stems when the NEG and VOL suffixes follow. In the Izumo City dialect, RG occurs in any stem when the NEG, VOL, and IMP suffixes follow. In this paper, these three conditions of the verb stems and suffixes in which RG occurs will be discussed.

According to my own and other previous descriptions of RG in each dialect, the type of verb stem and succeeding suffixes are involved in whether RG occurs or not. Therefore, my survey data comprised the full set of 20 specific verb forms (four verb stems × five inflectional suffixes), including negative evidence. In the formulations, it is shown that the hierarchical models mentioned in (1) and (2) are valid in order to capture the conditions of RG in each dialect focusing on the stem-final vowel, the stem mora count, and the succeeding suffixes.

⁴ Fertig (2013) classifies this type of change as an across-the-board change, in which patterns extend to entire classes of candidate forms, not lexically item-by-item.

3 A new approach to RG

3.1 The survey

The survey for this study differed from those used in previous studies in its incorporation of comprehensive data for each dialect. The data include descriptions of verb morphology from previous studies as well as my own survey data for some dialects. The objective of the survey was to examine dialect areas in which previous studies have reported RG to occur, and to determine whether or not each particular RG form is acceptable in each dialect. To make the data comparable, a standardized instrument with the survey targets shown in Table 4 was used.

Table 4: Survey targets

Stems			Suffixes			
vowel	morae	exam- ple	mood	tense	polarity	example
/i/	1	‘see’	indicative	non-past	affirmative	‘see’
/i/	2	‘get up’	indicative	non-past	negative	‘not see’
/e/	1	‘sleep’	indicative	past	affirmative	‘saw’
/e/	2	‘open’	volitional	---	affirmative	‘will see’
			imperative	---	affirmative	‘Look!’

First, for the verb stems, the variables were stem-final vowel (/i/ and /e/) and stem mora count (monomoraic and bimoraic). For each dialect, the word forms of four verb stems were examined: mono- and bimoraic stems with stem-final vowel /i/, and mono- and bimoraic stems with stem-final vowel /e/. Next, five suffixes were examined: affirmative non-past indicative (NPST), negative non-past indicative (NEG), affirmative past indicative (PST), affirmative volitional (VOL), and affirmative imperative (IMP).

There are some dialects in which RG occurs in trimoraic verb stems (e.g., /hazime-/ ‘start’), alternating stems (i.e., /ku-/ ~ /ko-/ ~ /ki-/ ‘come,’ and /su-/ ~ /se-/ ~ /si-/ ‘do’), compound verb stems (e.g., /nozok-i+mi-/ ‘peek at’), and auxiliary verb stems (e.g., /V-te mi-/ ‘try to V’). However, as previous studies provide very few descriptions of such cases, it is difficult to formulate them. Thus, they are not included in the scope of this study. For the same reason, negative past suffixes, suffixes forming the so-called *ren’yo* forms (e.g., polite suffix /-mas/), *ren’yo* forms following no suffix (e.g., so-called *ren’yo* imperative forms in Western dialects), and causative suffixes are also excluded from this study. These verb stems and suffixes are discussed in §6.

There are word forms such as *mijan* ‘do not see’ in southern Kinki dialects, and for these forms, it is suggested that the r > j sound change occurred in the RG form such as *miran* (e.g., Hidaka 2014). However, there is no synchronic or diachronic description of the alternation of

the stem-final consonant *r* to *j* or of the *j*-stem in such dialects, which does not correspond to RG in this paper and is therefore beyond its scope.

This paper surveys the conditions for RG in each dialect and establishes typological generalizations (e.g., Comrie 1989; Song 2001; Croft 2002). The framework of intragenetic typology (Kibric 1998; Daniel 2010) is adopted to formulate the microvariations in the conditions for RG in Japanese dialects as implicational universals (Rickford 2004; Corbett 2010), which are in turn expressed as hierarchies. Formulations as implicational universals require exhaustive data on the values of the variables under consideration (Rickford 2004). Therefore, when referring to descriptive data from previous studies, only those that show all 20 verb forms (four verb stems × five suffixes) in Table 4 are adopted.

There are two more reasons why some previous studies are not included in this paper. First, they present no negative evidence. As discussed in §2, the absence of negative evidence that the RG form is unacceptable when there are no RG forms precludes determining whether or not it contradicts the hierarchies. Second, these studies provide insufficient data for completing Table 4, and it is impossible to determine whether the missing word forms contradict the hierarchies. For example, as shown in Table 2, the GAJ is missing data for the monomoraic *i*-final ‘see’ volitional form. Thus, even if the GAJ data for one dialect are consistent with the hierarchies, it cannot be determined whether or not that data would remain consistent if it included the missing categories. Moreover, because RG is an ongoing change, it would be inappropriate to use data from the survey from the present study to fill in gaps in these other studies, for the current data may not accurately reflect what was true in the past.

3.2 Dialect types

Each dialect is classified into the dialect types shown in Table 5: Eastern-type, Western-type, and Kyushu-type. These three distinct types are identified according to the dialect area (Tojo 1953) and the morphological system of each dialect (Miyaoaka 2022). According to the dialect types, the attested patterns of verb stems and suffixes exhibiting RG differ as shown in §4.

Table 5: Dialect types

	dialect area	stem-final vowel alternation	/r/
Eastern-type	Eastern Japan	–	suffix-initial
Western-type	Western Japan	–	suffix-initial
Kyushu-type	mainly Kyushu	+	stem-final

The “Dialect area” indicates which of the dialect areas (Tojo 1953) of the Japanese dialect belongs to. This classification is introduced since the condition of RG differs depending on the dialect area. For example, in Eastern dialects, RG occurs only when the succeeding suffixes are imperative and/or volitional suffixes, not when they are negative non-past or past suffixes

(Hikosaka 2018). When the past suffix follows, RG occurs only in a few dialects, all of which are of the Kyushu type (Kobayashi 1996; Hikosaka 1999).

The stem-final vowel alternation column of Table 5 indicates whether or not there are vowel-final stem classes in which the final vowel alternates (i.e., in the Japanese linguistic term *nidan dōshi*) in the verb morphology of the dialect in question. In the dialects of Kyushu and those of the mountainous area of the southern Kinki region, there are vowel-final stems in which the final vowel alternates /i/ ~ /u/ or /e/ ~ /u/ (e.g., Okumura 1968; Kobayashi 1997; 2004). In these dialects, it is said that the /r/ of the RG forms is stem final (Miyaoka 2022) and that the conditions of the stems in which RG occurs are restricted (Kobayashi 1997; 2004; Onishi 2008; see §5.1).

The “/r/” column of Table 5 indicates whether /r/ in the RG forms occurs synchronically stem-finally or suffix-initially (Miyaoka 2022). In order to analyze whether the /r/ of RG forms belongs to either the stem-final position or suffix-initial position, Miyaoka (2022) describes the derivational and inflectional verb forms in two dialects where RG occurs. In the first dialect, the past, negative non-past, volitional, and imperative forms of /mi-/ ‘see’ are *mita* ‘saw’, *miran* ‘do not see’, *mira* ‘will see’, and *mire* ‘Look!’. In this dialect, the causative forms of /mi-/ ‘see’ are *miraseta* ‘made someone see’, *miraseran* ‘do not make someone see’, *mirasera* ‘will make someone see’, and *mirasere* ‘make someone see’. The /r/ can occur between the verb stem and the derivational causative suffix as well as between the causative derivational suffix and inflectional suffixes. The causative suffix can be analyzed as e-final based on the causative past forms *miraseta*. The causative forms are thus analyzed economically and uniformly as *miraseta* /mi-rase-ta/, *miraseran* /mi-rase-ran/, *mirasera* /mi-rase-ra/, and *mirasere* /mi-rase-re/. Following this analysis, the negative non-past, volitional, and imperative suffixes in this dialect are /-ran/, /-ra/, and /-re/, and the verb forms *miran*, *mira*, and *mire* in this dialect are analyzed in a parallel fashion as /mi-ran/, /mi-ra/, and /mi-re/. As such, the /r/ in these RG forms belongs suffix-initially in this dialect. On the contrary, in the second dialect, the past, negative non-past, volitional, and imperative forms of /mi-/ ‘see’ are *mita* ‘saw’, *miran* ‘do not see’, *miroo* ‘will see’, and *mire* ‘Look!’. In this dialect, the causative forms of /mi-/ ‘see’ are *miraseta* ‘made someone see’, *mirasen* ‘do not make someone see’, *mirasjuu* ‘will make someone see’, and *mirasjee* ‘make someone see’. The /r/ can occur only between the verb stem and the derivational causative suffix. The underlying representations of these forms are thus analyzed economically and uniformly as *mita* /mi-ta/, *miran* /mir-a-n/, *miroo* /mir-a-u/, *mire* /mir-e/, *miraseta* /mir-ase-ta/, *mirasen* /mir-ase-n/, *mirasjuu* /mir-ase-u/, and *mirasjee* /mir-ase-i/. As such, the /r/ of these RG form belongs stem-finally in this dialect. As indicated above, the position of the /r/ in RG forms differs from dialect to dialect. As mentioned in Footnote 1, the two are jointly referred to as RG as in previous studies. However, in this paper, the positions of the /r/ are distinguished in order to formulate the condition for RG. The former analysis is generally applicable to Western and

Eastern dialects, whereas the latter can be applied to almost all Kyushu dialects, based on the synchronic data in each dialect.⁵

3.3 Data sources

The present study examined 60 dialects (by dialect area, four in Tohoku, one in Hokuriku, two in Tokai, six in Kinki, one in Umpaku, six in Chugoku-Shikoku, and forty in the Kyushu area). The data sources for these dialects are shown in Figure 1 and Tables 6–8. In Figure 1, the broken-line circles indicate the areas where RG exists, and the dots indicate the locations of the dialects of the data sources. Even within the same location, verb morphology data may differ according to participant age or age group. In this paper, the different locations and age groups are each considered as one linguistic system, the conditions for RG in each system are identified, and the conditions for RG are formulated to explain all of the systems.⁶

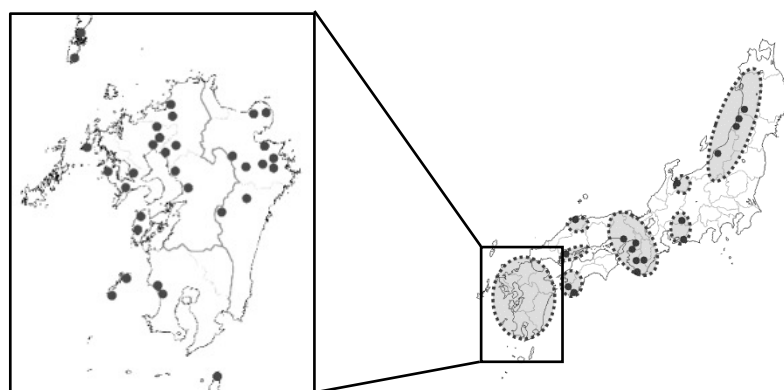


Figure 1: Geographical distribution of data of the dialects used in the hierarchies

⁵ Miyaoka (2022) points out that in Kyushu dialects, /r/ in RG forms belongs to the verb stem. However, in some Kyushu dialects, for imperative RG forms /r/ belongs to the suffix. Additionally, in some Kyushu dialects (Chikuzen, Chikugo (Fukuoka Pref.), Iki, Tsushima, and Goto Islands (Nagasaki Pref.)), the imperative form of the verb stem ‘do’ is *sere*. These dialects are indicated with an asterisk (*) in Table 8. In addition to the word form *sere*, the forms of the causative imperative in these dialects are *mirasere* ‘make someone see it’ (Kurume City dialect; Sakata 1986, Iki Island dialect; Yamaguchi 1937), *kerasere* ‘make someone kick it’ (Goto Island dialect; Kadoya 2020). Thus, the underlying representation of *sere* is /se-re/ and that of the causative imperative forms is /X-ase-re/. Like these, in such dialects, the imperative suffix is set to /-re/, in which case /r/ in imperative RG forms (e.g., *mire* ‘Look!’) belongs to the suffix (i.e., *mire* /mi-re/). The hierarchies in this paper also cover these dialects. However, for these dialects, the imperative suffixes are not considered in the suffix hierarchies, as the morphological structure of the imperative RG forms differs from the other word forms.

⁶ The surveys of these dialects were conducted between 2019 and 2022. The birth years of the participants were as follows. Sakata (Yamagata Pref.): 1943–1947; Ena (Gifu Pref.): 1943–1951; Kishiwada (Osaka Pref.): 1958; Miyama (Wakayama Pref.): 1951; Izumo (Shimane Pref.): 1950; Amakusa (Kumamoto Pref.): 1945–1953; Shiba (Miyazaki Pref.): 1926–1933; Ichiki-Kushikino (Kagoshima Pref.): 1942. Every participant had lived in his or her respective location from infancy and until at least age 18.

Table 6: Primary sources of hierarchies and patterns of verb stems and suffixes in Eastern-type dialects

prefecture	dialects	sources	stem	suffix			
				/i/ 1	/i/ 2	/e/ 1	/e/ 2
Akita	Former Chōkai Town	Matsumaru (2002c; 2002d)	ε	E2	E2	E2	E2
Yamagata	Sakata City	author data	ε	E2	E2	E2	E2
Yamagata	Ōyama, Tsuruoka City	Onishi (1994)	ε	E2	E2	E2	E2
Nīgata	Former Sakae Village	Kato (1961)	ε	E2	E2	E2	E2

Table 7: Primary sources of hierarchies and patterns of verb stems and suffixes in Western-type dialects

prefecture	dialects	sources	stem	suffix			
				/i/ 1	/i/ 2	/e/ 1	/e/ 2
Toyama	Toyama City	Konishi (2016)	ε	W2	W2	W2	W2
Gifu	Ena City	author data	ε	W3	W3	W3	W3
Shizuoka	Former Arai Town	Yamaguchi (1985)	ε	W3	W3	W3	W3
Hyōgo	Akashi City	Tachibana (1966)	ε	W3	W3	W3	W3
Ōsaka	Ōsaka City	Wada (1961)	ε	W1	W1	W1	W1
Ōsaka	Kishiwada City	author data	ε	W3	W3	W3	W3
Shimane	Izumo City	author data	ε	W3	W3	W3	W3
Hiroshima	Etajima City	author data	ε	W3	W2	W3	W2
Ehime	Ainan Town	author data	ε	W1	W1	W1	W1
Kōchi	Sukumo City: 3 age groups	Matsumaru (2001)	ε	W2	W2	W2	W2
Kōchi	Hata County	Hamada (1982)	ε	W3	W3	W3	W3

Table 8: Primary sources of hierarchies and patterns of verb stems and suffixes in Kyushu-type dialects

prefecture	dialects	sources	stem	suffix			
				/i/ 1	/i/ 2	/e/ 1	/e/ 2
Nara	Totsukawa Village	Hirayama (1979)	ε	K2	K2	K2	K2
Wakayama	Miyama, Hidakagawa Town	author data	ε	K3	K1	K3	K1
Wakayama	Tanabe City	Hirasawa (1982)	ε	K2	K2	K2	K1
Fukuoka	Hakata, Fukuoka City: Senior*	Hayata (1985)	γ	K3	K0	K3	K0
Fukuoka	Hakata, Fukuoka City: Younger*	Jinnouchi (1989)	ε	K3	K3	K3	K3
Fukuoka	Former Shake Town, Fukuoka City*	Okano (1991)	ε	K3	K3	K3	K1
Fukuoka	Kurume City	Sakata (1986)	δ	K3	K4	K3	K0
Fukuoka	Yanagawa City*	Matsuoka (2021)	δ	K3	K3	K3	K0
Saga	Kitayama, former Fuji Town	Iwamoto (1991)	δ	K3	K3	K3	K0
Saga	Saga City	Takayama (2016)	δ	K3	K3	K2	K0
Nagasaki	Itanoura, former Ōseto Town	Nobayashi (1966)	α	K3	K0	K0	K0
Nagasaki	Ōmura City	Shinozaki (1966)	ε	K3	K3	K3	K1
Nagasaki	Nagasaki City	Tsukamoto (1978)	β	K3	K3	K0	K0
Nagasaki	Kuta, former Izuhara Town*	Nobayashi (1966)	γ	K3	K0	K3	K0
Nagasaki	Toyotama, Tsushima City*	Harada (2006)	ε	K3	K1	K3	K1
Kumamoto	Agemasunaga, Arao City*	Nobayashi (1966)	β	K3	K4	K0	K0
Kumamoto	Kikuchi City	Fujimoto (2002; 2004)	δ	K3	K4	K3	K0
Kumamoto	Imatomi, former Kawaura Town	Matsumaru (2002a; 2002b)	δ	K3	K3	K1	K0
Kumamoto	Reihoku Town	Kuroki (2019)	β	K3	K3	K0	K0
Kumamoto	Ushibuka, Amakusa City	author data	δ	K3	K3	K3	K0
Ōita	Himeshima Village	Tane (1983)	ε	K2	K1	K2	K1
Ōita	Kataku, Kunisaki Town: Senior, middle	Tane (1983)	ε	K3	K1	K3	K1
Ōita	Kataku, Kunisaki Town: Younger	Tane (1983)	ε	K3	K2	K3	K1
Ōita	Kunisaki	Kuribayashi (2009)	γ	K3	K0	K3	K0
Ōita	Dairiki, Bungotakada City	Oishi (1996)	γ	K2	K0	K2	K0
Ōita	Kokonoe Town	Itoi (1964)	γ	K2	K0	K2	K0
Ōita	Former Kawato Village	Itoi (1959)	γ	K2	-	K2	K0
Ōita	Nishikono, former Notsu Town	Itoi (1960)	γ	K2	-	K2	K0
Ōita	Nagayu, former Naoiri Town	Nobayashi (1991)	γ	K2	-	K2	K0
Ōita	Furugo, Oita City	Itoi (1991)	γ	K2	-	K2	K0
Ōita	Motomachi, Oita City	Itoi (1983)	γ	K2	-	K2	K0
Ōita	Usuki City	Hayata (1981)	γ	K2	K0	K2	K0
Miyazaki	Former Hinokage Town	Nomoto (1959)	δ	K4	K4	K4	K0
Miyazaki	Shiba Village	author data	δ	K3	K3	K3	K0
Kagoshima	Kagoshima City	Goto (1994)	δ	K4	K3	K3	K0
Kagoshima	Ichiki-Kushikino City	author data	δ	K3 K4	K4	K3 K4	K0
Kagoshima	Former Sato Village	Kuroki (2019)	δ	K3	K4	K3	K0
Kagoshima	Teuchi, former Shimokoshiki Village	Kuroki (2019)	γ	K3	K4	K0	K0
Kagoshima	Nishinoomote, Nishinoomote City	Kamimura (1959)	δ	K3	K4	K3	K0
Kagoshima	Tanegashima Island	Uemura (1970)	δ	K3	K4	K3	K0
Kagoshima	Former Issō, Kamiyaku Town	Uemura (1968)	δ	K4	K4	K1	K0

4 Implicational hierarchies

4.1 Hierarchies for verb stems

As shown in §3.1, this study investigated four verb stems, resulting in 15 logically possible patterns of RG (i.e., $2^4 - 1$, excluding one in which RG does not occur in any verb stems). Of these 15 possible patterns, only five are attested in the actual data, as shown in Table 9, where the columns indicate the stem-final vowel and the rows indicate the stem mora count. The ‘+’ and ‘-’ indicate whether RG occurs or does not occur, respectively. In Eastern- and Western-type dialects, only one pattern (ϵ) in Table 9 actually occurs, whereas Kyushu-type dialects all five patterns.⁷

Table 9: Verb stem patterns

Pattern α			Pattern β			Pattern γ		
1	2		1	2		1	2	
/i/	+	-	/i/	+	+	/i/	+	-
/e/	-	-	/e/	-	-	/e/	+	-

Pattern δ			Pattern ϵ		
1	2		1	2	
/i/	+	+	/i/	+	+
/e/	+	-	/e/	+	+

Tables 6–8 include columns indicating which pattern in Table 9 each dialect exhibits. The following sets of examples illustrate each pattern in the various different dialects.

(5) Pattern α : Itanoura, former Oseto Town (Nagasaki Pref.; Nobayashi 1966)

a. <i>mir-an</i>	b. <i>oki-n</i>	c. <i>ne-n</i>	d. <i>uke-n</i>
see-NEG	get.up-NEG	sleep-NEG	receive-NEG
‘do not see’	‘do not get up’	‘do not sleep’	‘do not receive’

⁷ Some data in Eastern- and Western-type dialects show a difference in verb morphology depending on stem mora count and stem-final vowel. In (I) Kishiwada City (author data), RG occurs in monomoraic stems when a *ren’yō* suffix follows, although it does not occur in bimoraic stems. In (II) Etajima City (author data), RG occurs in monomoraic stems when the negative non-past suffix follows, although it does not occur in bimoraic stems. In (III) Takasago City (Hara 1956, not considered in this paper for its lack of exhaustive data), while data on e-final bimoraic stems is lacking, RG occurs in i-final stems when the imperative suffix follows, although it does not occur in e-final stems.

(I) a. <i>miritai</i> ‘want to see’	b. <i>okitai</i> ‘want to get up’	c. <i>neritai</i> ‘want to sleep’	d. <i>aketai</i> ‘want to open’
(II) a. <i>miran</i> ‘not see’	b. <i>okin</i> ‘not get up’	c. <i>neran</i> ‘not sleep’	d. <i>aken</i> ‘not open’
(III) a. <i>mire</i> ‘look!’	b. <i>otire</i> ‘fall!’	c. <i>nee</i> ‘sleep!’	d. <i>kasane</i> ‘repeat!’

The stem patterns mean that RG may or may not occur after any suffix, not any particular one. Following this schema, the dialects shown above are ϵ -type, since RG occurs in all vowel-final stems after any suffix (e.g., volitional). Upon consideration of these data, however, the stem mora count and the stem-final vowel both affect RG conditions in Eastern- and Western-type dialects.

- (6) Pattern β: Nagasaki City (Nagasaki Pref.; Tsukamoto 1978)
- | | | | |
|------------------|-------------------|-----------------|-----------------|
| a. <i>kir-an</i> | b. <i>okir-an</i> | c. <i>de-n</i> | d. <i>age-n</i> |
| wear-NEG | get.up-NEG | go.out-NEG | raise-NEG |
| ‘do not wear’ | ‘do not get up’ | ‘do not go out’ | ‘do not raise’ |
- (7) Pattern γ: Kokonoe Town (Oita Pref.; Itoi 1964)
- | | | | |
|------------------|-----------------|------------------|------------------|
| a. <i>mir-an</i> | b. <i>oki-n</i> | c. <i>ner-an</i> | d. <i>uke-n</i> |
| see-NEG | get.up-NEG | sleep-NEG | receive-NEG |
| ‘do not see’ | ‘do not get up’ | ‘do not sleep’ | ‘do not receive’ |
- (8) Pattern δ: Shiba Village (Miyazaki Pref.; author data)
- | | | | |
|------------------|-------------------|------------------|------------------|
| a. <i>mir-an</i> | b. <i>okir-an</i> | c. <i>ner-an</i> | d. <i>hute-n</i> |
| see-NEG | get.up-NEG | sleep-NEG | discard-NEG |
| ‘do not see’ | ‘do not get up’ | ‘do not sleep’ | ‘do not discard’ |
- (9) Pattern ε: Totsukawa Village (Nara Pref.; Hirayama 1979)
- | | | | |
|------------------|-------------------|------------------|-------------------|
| a. <i>mir-an</i> | b. <i>okir-an</i> | c. <i>ner-an</i> | d. <i>uker-an</i> |
| see-NEG | get.up-NEG | sleep-NEG | receive-NEG |
| ‘do not see’ | ‘do not get up’ | ‘do not sleep’ | ‘do not receive’ |

The attested patterns allowed us to make implicational hierarchies summarized in (1).

- a. the stem-final vowel: /i/ > /e/
- b. stem mora count: 1 > 2

According to (1a), other things being equal (i.e., the number of morae is the same), if RG occurs in a stem with the stem-final vowel /e/, then it also occurs in a stem with the stem-final vowel /i/. According to (1b), other things being equal (i.e., the stem-final vowel is the same), if RG occurs in a bimoraic stem, then it also occurs in a monomoraic stem. In the stem-final vowel hierarchy, i/u-stem corresponds to the final vowel /i/, and e/u-stem corresponds to the final vowel /e/ because, when these stems are followed by a suffix that is involved in RG (i.e., negative non-past, past, volitional, imperative, etc.), the final vowel in the i/u-stem and the e/u-stem is /i/ and /e/, respectively.

With these implicational hierarchies, it is claimed that the non-attested patterns (i.e., 10 out of the possible 15) are not accidental gaps but are ruled out since they violate the hierarchy. In other words, the patterns shown in Table 10 are not attested and are thus excluded from the hierarchies.

Table 10: Patterns NOT attested and thus ruled out in the hierarchies

1 2	1 2	1 2	1 2	1 2
/i/ + -	/i/ + +	/i/ + -	/i/ - +	/i/ - -
/e/ - +	/e/ - +	/e/ + +	/e/ - -	/e/ + -
1 2	1 2	1 2	1 2	1 2
/i/ - -	/i/ - +	/i/ - +	/i/ - -	/i/ - +
/e/ - +	/e/ + -	/e/ - +	/e/ + +	/e/ + +

4.2 Hierarchies for suffixes

A closer look at the data introduces another factor that is independent of those pertaining to the stem (i.e., stem-final vowel and stem mora count), namely one involving the inflectional suffixes. Not only the stems but also the inflectional suffixes, another major component of verb inflection, play an important role in RG.

For four inflectional suffixes (i.e., imperative (IMP), volitional (VOL), negative (NEG), and past (PST)), a clear pattern emerges regarding whether or not RG can occur in different dialects. Namely, the likelihood of RG varies according to the kind of suffix that is attached to a verb stem which potentially allows RG. Generally speaking, other things being equal (i.e., if the stem has the same mora length and ends in the same vowel), the irrealis-oriented suffixes (IMP, VOL, and NEG) are more likely to allow RG than the realis-oriented suffix (PST). Thus, if a dialect allows RG with a particular stem, then the next question is which suffix the stem takes and whether the suffix that is most likely to trigger RG is one of IMP, VOL, or NEG. If RG is disallowed for these suffixes, then RG is never allowed for the realis-oriented PST suffix.

The likelihood of RG differs among the irrealis-oriented suffixes (IMP, VOL, and NEG). In order to make a precise and robust formulation, it is necessary to consider different dialect types (Eastern, Western, and Kyushu) separately. As shown in §3.1, this paper investigated four suffixes, yielding 16 (i.e., 2^4) logically possible patterns in which RG may or may not occur. In Eastern-type dialects, only one pattern actually occurs (see Table 11). In Western-type dialects, only three patterns actually occur (see Table 12). In Kyushu-type dialects, only five patterns actually occur (see Table 13).

Table 11: Suffix patterns in Eastern-type dialects

pattern	IMP	VOL	NEG	PST
E2	+	+	-	-

Table 12: Suffix patterns in Western-type dialects

pattern	VOL	IMP	NEG	PST
W1	+	-	-	-
W2	+	+	-	-
W3	+	+	+	-

Table 13: Suffix patterns in Kyushu-type dialects

pattern	VOL	NEG	IMP	PST
K0	-	-	-	-
K1	+	-	-	-
K2	+	+	-	-
K3	+	+	+	-
K4	+	+	+	+

Tables 6–8 show which pattern from Tables 11–13 each dialect has. The following sets of examples illustrate each pattern from the various dialects.

(10) Pattern E2: Sakata City (Yamagata Pref.; author data); monomoraic i-final stem

a. <i>mi-re</i> see-IMP 'Look!'	b. <i>mi-ro</i> see-VOL 'will see'	c. <i>mi-ne</i> see-NEG 'do not see'	d. <i>mi-ta</i> see-PST 'saw'
---------------------------------------	--	--	-------------------------------------

(11) Pattern W1: Osaka City (Osaka Pref.; Wada 1961); monomoraic e-final stem

a. <i>ne-roo</i> sleep-VOL 'will sleep'	b. <i>ne-e</i> sleep-IMP 'Sleep!'	c. <i>ne-n</i> sleep-NEG 'do not sleep'	d. <i>ne-ta</i> sleep-PST 'slept'
---	---	---	---

(12) Pattern W2: Toyama City (Toyama Pref.; Konishi 2016); bimoraic i-final stem

a. <i>oki-roo</i> get.up-VOL 'will get up'	b. <i>oki-re</i> get.up-IMP 'Get up!'	c. <i>oki-n</i> get.up-NEG 'do not get up'	d. <i>oki-ta</i> get.up-PST 'got up'
--	---	--	--

(13) Pattern W3: Arai Town (Shizuoka Pref.; Yamaguchi 1985); bimoraic e-final stem

a. <i>tabe-roo</i> eat-VOL 'will eat'	b. <i>tabe-re</i> eat-IMP 'Eat!'	c. <i>tabe-ran</i> eat-NEG 'do not eat'	d. <i>tabe-ta</i> eat-PST 'ate'
---	--	---	---------------------------------------

(14) Pattern K0: Ushibuka, Amakusa City (Kumamoto Pref.; author data); bimoraic e-final stem

a. <i>akjuu</i> open.VOL 'will open'	b. <i>ake-n</i> open-NEG 'do not open'	c. <i>ake-ro</i> open-IMP 'Open!'	d. <i>ake-ta</i> open-PST 'opened'
--	--	---	--

(15) Pattern K1: Imatomi (Kumamoto Pref.; Matsumaru 2002a; 2002b); monomoraic e-final stem

a. <i>ner-oo</i> sleep-VOL 'will sleep'	b. <i>ne-n</i> sleep-NEG 'do not sleep'	c. <i>ne-ro</i> sleep-IMP 'Sleep!'	d. <i>ne-ta</i> sleep-PST 'slept'
---	---	--	---

(16) Pattern K2: Kokonoe Town (Oita Pref.; Itoi 1964); monomoraic i-final stem

a. <i>mir-oo</i> see-VOL 'will see'	b. <i>mir-an</i> see-NEG 'do not see'	c. <i>mi-jo</i> see-IMP 'Look!'	d. <i>mi-ta</i> see-PST 'saw'
---	---	---------------------------------------	-------------------------------------

(17) Pattern K3: Omae village (Miyazaki Pref.; author data); bimoraic i-final stem

a. <i>okir-oo</i> get.up-VOL 'will get up'	b. <i>okir-an</i> get.up-NEG 'do not get up'	c. <i>okir-e</i> get.up-IMP 'Get up!'	d. <i>oki-ta</i> get.up-PST 'got up'
--	--	---	--

(18) Pattern K4: former Hinokage Town (Miyazaki Pref.; Nomoto 1959); bimoraic i-final stem

a. <i>okir-oo</i> get.up-VOL 'will get up'	b. <i>okir-an</i> get.up-NEG 'do not get up'	c. <i>okir-e</i> get.up-IMP 'Get up!'	d. <i>okit-ta</i> get.up-PST 'got up'
--	--	---	---

In each of the three dialect types, under the condition that the stem factors are controlled (i.e., the stem-final vowel and the stem mora count are the same), if RG occurs in a certain suffix of the hierarchy, then RG also occurs in suffixes to the left of it summarized in (2):

Volitional > Negative non-past > Past

- Eastern-type: Imperative = Volitional
- Western-type: Volitional > Imperative > Negative non-past
- Kyushu-type: Volitional > Negative non-past > Imperative > Past

Note that (2) holds under the condition that the stem factors are controlled (i.e., fixed for a particular combination of factors like monomoraic + i-final). In fact, there is an interaction between the stem factors and the suffix factor. For example, the Kokonoe Town dialect (Oita Prefecture; Itoi 1964) has pattern K2 if the stem is monomoraic, whereas it has pattern K0 if the stem is bimoraic.

The equal sign ‘=’ indicates a lack of any hierarchical relationship according to the data surveyed in this paper. In other words, in Eastern-type dialects, there are data showing that RG occurs when both imperative and volitional suffixes follow. There are no data showing that it occurs only when the imperative suffix follows or only when the volitional suffix follows.⁸ Exhaustive data from other dialects should be investigated to determine whether there is a preferred order, which is a remaining issue.

5 Theoretical account of the value ordering of the hierarchies: From the viewpoint of a usage-based theory

In this section, I will present a theoretical interpretation of why the values in the hierarchies of the verb stems (§4.1) and the suffixes (§4.2) are in the observed order. A review of previous studies (§5.1) will show that the hierarchies can be explained in a unified way in terms of frequency (§5.2).

5.1 Previous studies

Regarding the hierarchies of the verb stems (1), it has been pointed out in previous studies that RG does not occur when the final vowel of the stem alternates. Kobayashi (1996), Onishi (2008), and de Chene (2020) argue as follows. In relation to monograde-shift (*ichidanka* in Japanese linguistic terms), there are dialects in Kyushu where bimoraic stems are i/u- and e/u-stems with alternating stem-final vowels (*nidan* ‘bigrade verb stems’ in Japanese linguistic terms), and in many of these, RG does not occur.⁹ Thus, the stems undergoing RG must be i- and e-stems (i.e., stems in which monograde-shift has occurred). Onishi (2008) explains that the reason for this constraint is that there is an order in which i/u- and e/u-stems change to i-stems and e-stems through conjugational rationalization and that these stems change to r-final stems (i.e., RG stems) by analogical change.

⁸ As an anonymous reviewer pointed out, there are some data in the GAJ in which RG occurs in (I) imperative forms and does not occur in (II) volitional forms. As an example, the GAJ data from Kashiwazaki City, Niigata Prefecture, are shown below.

(I)	a. <i>mire</i> ‘look!’	b. <i>okire</i> ‘get up!’	c. ND	d. <i>akere</i> ‘open!’
(II)	a. ND	b. <i>okijoo</i> ‘will get up’	c. <i>nejoo</i> ‘will sleep’	d. <i>akejoo</i> ‘will open’

However, as discussed in §2 and §3.1, the GAJ data are neither exhaustive, nor do they provide any negative evidence. For this reason, the GAJ data do not allow us to identify the preferred order of imperative and volitional suffixes.

⁹ There are some data in which RG occurs even in e/u-stems (e.g., *nuru* /nu-ru/ ‘sleep,’ *neta* /ne-ta/ ‘slept,’ *neran* /ner-a-n/ ‘do not sleep’ in Yanagawa City dialect, Fukuoka Prefecture; Matsuoka 2021).

In addition, the conditions of the vowel-final stem alternation (i.e., /i/ ~ /u/ in i/u stems, and /e/ ~ /u/ in e/u stems) in verb stems are also explained by the stem mora count and the stem-final vowel (Okumura 1968 and according to my data). Fundamentally, we need to explain the hierarchies of the verb stems presented in the current paper.

Regarding the hierarchies of the suffixes (2), Kobayashi (1996) and Sasaki (2019) argue that the variance in the likelihood of RG occurrence is due to the succeeding suffixes, which involve different motivation for the change. Kobayashi (1996) argues the reasons suffix by suffix. Sasaki (2019) also discusses the suffix-specific reasons and explains that the likelihood of RG differs suffix by suffix because the likelihood of the change that happens in each suffix differs. Sasaki (2019) claims that the change in grammar (e.g., /r/ insertion rule in volitional suffix) tends to occur more often than RG with lexical item changes (e.g., negative non-past and imperative suffixes).

The present paper explains the value ordering of the hierarchies of both verb stems (1) and their suffixes (2) in a unified way. As mentioned above, previous works analyzed the conditions of stems and suffixes separately. To begin with, the methodology for deriving implicational hierarchies (1) and (2) in this paper differs from that of previous works in that the grammatical systems differ from dialect to dialect and that we formulate the conditions of RG uniquely for each dialect (§3). Next, I will provide a unified and empirical explanation of all the hierarchies of RG (1) and (2) by introducing the perspective of the frequency. This explanation does not contradict the analyses in previous work mentioned above.

5.2 Frequency

In this section, I will explain the order of the values of each variable in the hierarchies (1) and (2) by focusing on the frequency of word forms. From the viewpoint of usage-based theory (e.g., Hooper 1976; Bybee 1985; Phillips 2001), analogical change is more likely to occur in infrequently used word forms than in frequently used ones. For example, as Bybee (1985) showed, an infrequently used strong verb class in Middle English (e.g., *mow-mew*) is regularly conjugated in Modern English by analogical change with a weak verb class (i.e., *mow-mowed*). On the other hand, their high-frequency counterparts (e.g., *know-knew*) do not undergo analogical change and remain irregular in conjugation. The reason for this difference is that high-frequency words are memorized as they are and thus less likely to undergo the type of analysis that leads to analogical change.

RG is considered to be an analogical change from a vowel-final stem to an r-stem or from a vowel-initial suffix to an r-initial suffix (see footnote 1). Thus, its occurrence should be more likely in less frequently used word forms. Accordingly, the hypothesis in this paper is that the hierarchies (1) and (2) reflect the frequency of word forms. Next I will show that the hierarchies in (1) and (2) are ordered from left to right in order of increasing frequency and argue that the hypothesis is indeed valid.

The survey for this study used the Corpus of Japanese Dialects (COJADS) of the National Institute for Japanese Language and Linguistics. As of April 2022, when the survey was conducted, COJADS (Ver. 2022.03) contained approximately 80 hours of discourse data from the dialects of 80 locations (not including Ryukyuan languages) from the Urgent Survey of Dialects conducted from 1977 to 1985. The survey covered approximately 30 hours of data from 19 locations (by

dialect area, two in Tohoku, one in Hokuriku, one in Tokai, four in Kinki, one in Umpaku, and 10 in Kyushu), where RG forms had previously been reported. The frequency of word forms consisting of the verb stems and suffixes shown in Table 4 were examined, and the specific search method in COJADS is shown in Table 14.¹⁰

Table 14: The COJADS search method

vowel	/i/	part of speech ‘verb’, conjugation type ‘upper monograde’
	/e/	part of speech ‘verb’, conjugation type ‘lower monograde’
morae	1, 2	confirmed and classified by the author
suffix	VOL	part of speech ‘verb’, conjugation form ‘volitional and inference’
	NEG	part of speech ‘verb’, conjugation form ‘mizen,’ following lexeme ‘nai (negative non-past)’
	PST	part of speech ‘verb’, conjugation form ‘ren’yo,’ following lexeme ‘ta (past)’

In the survey, the frequency of word forms was counted according to the method used in previous studies discussing frequency and analogical change. First, we listed the tokens of verb forms consisting of certain verb stems and suffixes for each word form. Second, we calculated the type and token frequencies of verb stems and the token frequency of suffixes following Matsuda (1989) and Koyanagi (2022), which examine the motivations for analogical change in classical and modern Japanese using the frequency of verb stems and suffixes. Because the token frequencies of verb stems differ greatly from one lexical element to another, we followed Bybee (1985) and Koyanagi (2022) and calculated the average token frequency for each verb stem by dividing the sum of the token frequencies for each stem class. As for the hierarchies of the suffixes, because the values and order of the hierarchies (2a–c) differ by dialect type, I will now discuss the order of the values of (2), which obtains for all dialects.

The frequency of verb stems and suffixes for the 19 dialects is shown in Tables 15 and 16 is divided according to stem-final vowel (/i/ or /e/) and stem mora count (monomoraic or bimoraic). The relationship between the frequency and the conditions of RG counted for each dialect was similar for all dialects.

First, we calculated frequencies for each of the categories of the verb stems (verb stem-final vowels: /i/ and /e/; stem mora count: 1 and 2), which were used as the values of the hierarchies (1). The hierarchy of the stem-final vowel (1a) takes /i/ and /e/ as values. The type frequency of i-final stems is 20 (= 6+14), the sum of their token frequency is 913 (= 431 + 482), and their average is 45.65 (913 ÷ 20). The type frequency of e-final stems is 97 (= 4 + 93), the sum of their token frequency is 2,850 (= 533 + 2,317), and their average is 29.38 (2,850 ÷ 97). The hierarchy of

¹⁰ I excluded the auxiliary verbs (e.g., /V-te mi-/ ‘try to V’) and compounded verbs (e.g., /nozok-i+mi-/ ‘peek at’) from the frequency calculations. Because the description in §3.3 covers main verbs with verb predicate endings and simple verbs with a single verb stem, the survey in §5 is also conditioned on these criteria. In addition, there are dialects in which the morphology of main verbs and auxiliary verbs differs. For example, in Yanagawa City dialect of the younger generation (Fukuoka Prefecture; author data), the negative non-past form of the main verb /mi-/ ‘to see’ is *miran*, **min*, and the negative non-past form of auxiliary verb /mi-/ ‘try to’ is *miran*, *min*. For this reason, I distinguish main verbs from auxiliary verbs.

the stem mora count (1b) takes 1 and 2 as values. The type frequency of monomoraic stems is 10 (= 6 + 4), the sum of their token frequency is 964 (= 431 + 533), and their average is 96.4 (96.4 ÷ 10). The type frequency of bimoraic stems is 107 (= 14 + 93), the sum of their token frequency is 2,799 (= 482 + 2,317), and their average is 26.15 (2,799 ÷ 107).

Table 15: Verb stem frequencies

vowel	morae	stem	freq.	vowel	morae	stem	freq.
/i/	1	mi- ‘see’	310	/e/	1	de- ‘go out’	440
/i/	1	ki- ‘wear’	79	/e/	1	ne- ‘sleep’	60
/i/	1	ni- ‘simmer’	19	/e/	1	e- ‘exist’	30
/i/	1	ni- ‘resemble’	12	/e/	1	e- ‘get’	3
/i/	1	i- ‘exist’	9				
/i/	1	hi- ‘dry’	2				
			sum (6 stems)				533
			431				
vowel	morae	stem	freq.	vowel	morae	stem	freq.
/i/	2	deki- ‘can’	242	/e/	2	tabe- ‘eat’	251
/i/	2	oti- ‘fall’	65	/e/	2	ire- ‘put in’	207
/i/	2	sugi- ‘pass’	39	/e/	2	tuke- ‘put’	179
/i/	2	kari- ‘borrow’	36	/e/	2	deke- ‘can’	176
/i/	2	oki- ‘get up’	25	/e/	2	age- ‘raise’	113
/i/	2	ori- ‘go down’	24	/e/	2	kake- ‘hang’	93
/i/	2	iki- ‘live’	16	/e/	2	ue- ‘plant’	70
/i/	2	tari- ‘be enough’	10	/e/	2	tome- ‘stop’	65
/i/	2	other 6 stems	25	/e/	2	other 85 stems	1,175
			sum (14 stems)				2,317
			482				

Table 16: Suffix token frequencies

VOL	NEG	PST
266	2,380	5,558

Table 17: Frequency by final vowel and stem mora count

	/i/	/e/	monomoraic	bimoraic
type	20	97	10	107
token (sum)	913	2,850	964	2,799
token (average)	45.65	29.38	96.4	26.15

Next, we confirmed the token frequencies of the categories of the suffixes. The hierarchies of the suffixes (2) take volitional, negative non-past, and past as values. The token frequencies of the suffixes are 266 for the volitional, 2,380 for the negative non-past, and 5,558 for the past.

According to the frequency numbers, the stem-final vowels /i/ and /e/ are in increasing order of type frequency. In terms of stem mora count types, monomoraic stems were less frequent than bimoraic ones. Thus, the value order of the hierarchies of the verb stems corresponds to the increasing order of the type frequency of the stems. It cannot be explained by the average token frequency based on the account of usage-based theory. The succeeding suffixes are, in increasing order of token frequency, volitional, negative non-past, and past. Thus, the value order of the hierarchies of the suffixes is from the bottom to the top of the hierarchy in order of increasing token frequency. There were no dialectal differences in token frequency in the same dialect area, and the same was true for the results for each dialect. Compared to the hierarchies (1) and (2), this finding is consistent with the prediction that infrequently used word forms are more likely to produce analogical change. Thus, it can be concluded that the hierarchies of verb stems (1) reflect the type frequency of the verb stems and that those of suffixes (2) reflect the token frequency of the suffixes.

6 Conclusion and remaining issues

This paper presented an analysis of a sample of 60 dialects featuring RG. The conditions for RG in the sampled dialects were formulated cross-dialectally in terms of stem-final vowel, stem mora count, and suffixes as shown by the implicational hierarchies in (1) and (2). These hierarchies, which were derived inductively, reflect the usage frequency in the corpus.

Nevertheless, a few issues remain. First, the verb stems and suffixes not covered in this paper need to be surveyed. This paper does not cover irregular verb stems (e.g., ‘come’ and ‘do’), compound verb stems, or stems that consist of three or more morae, nor does it cover causative, negative past, or *ren’yo* suffixes (see §3.1). The data collection instrument represented in Table 4 should also be used to examine the conditions for RG focusing on these verb stems and suffixes in each dialect, and the hierarchies presented in this paper should then be verified and expanded¹¹. Second, RG data need to be collected for other dialects. As mentioned above, the database for this paper was limited. In addition, comprehensive data from other dialects

¹¹ Within the limits of previous studies and my own data, it is tentatively possible to formulate the conditions of RG covering these verb stems and suffixes as follows:

- (I) Extended hierarchies of verb stem (temporal)
 - a. stem-final vowel: /i/ > /e/ > irregular (‘come’ and ‘do’)
 - b. stem mora count: 1 > 2 = 3
- (II) Extended hierarchies of affixes (temporal)
 - a. Eastern-type: Causative = Imperative = Volitional
 - b. Western-type: Causative = Volitional > Imperative > negative past > negative non-past > ren’yo
 - c. Kyushu-type: Causative = Volitional > Negative > imperative > ren’yo > past

Related to (Iib), most stems with four or more morae are compound verbs such as V_1+V_2 . In my own data, compound verbs behave morphologically the same as the verb stem of V_2 alone. Therefore, the conditions for RG for four or more morae can be considered the same as for the V_2 alone and to follow the hierarchies presented in this paper.

could help determine the hierarchical relationship of the values connected by the equal sign in the hierarchy presented in this paper. Moreover, such data might provide counterexamples, suggesting that these hierarchies should merely be interpreted as tendencies. In short, to verify and elaborate the hierarchies in this paper, further research on the morphological systems in RG dialects should be conducted.

Abbreviations

–: affix boundary / IMP: imperative / NEG: negative non-past / PST: past / RG: *ragyō-godanka* / VOL: volitional

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